# SIT Policy Advisor: A Reliable RAG Application

Live document-based QA with transparent citations and hallucination control

Presented by: [GOD"S FAVOURITE]



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THE PROBLEM TO SOLVE

# Students need answers they can trust

Policy answers need to be fast enough for real questions, precise enough for official guidance, and transparent enough to verify.



## The challenge

Official policy documents are dense, fragmented, and hard to search quickly.



## The solution

Local embeddings locate relevant passages; Gemini formulates precise answers from those passages.



## Core constraints

Document-grounded facts; exact page and document citations; refusal for out-of-domain questions.

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SYSTEM FLOW

# Five stages turn PDFs into cited answers

## Ingest

Read PDFs dynamically with pypdf.

## Chunk

Split pages into 120-word segments with 30-word overlap.

## Embed

Generate dense vectors locally with sentence-transformers/all-MiniLM-L6-v2.

## Retrieve

Use cosine similarity to select the top 3 passages.

## Generate

Send context plus strict prompt rules to gemma-4-26b-a4b-it.

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QUALITY CONTROLS

# Each hyperparameter protects answer quality

## 120

### 120-word chunks

Capture one dense policy rule or schedule event without excess noise.

## 30

### 30-word overlap

Keeps dates and conditions intact across sentence boundaries.

## 3

### Top-K = 3

Supplies enough background while limiting prompt bloat and response time.

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AUDITABLE BY DESIGN

# The live app makes every answer auditable

### Live citations

Factual claims carry explicit markers such as [SIT_Student_Handbook_and_Exam_Regulations.pdf, p. 4].

### Gradio UI

Multi-PDF upload, quick example questions, and an interactive Ask action.

### Transparency accordion

Exposes similarity scores and verbatim snippets so users can verify the source manually.

ILLUSTRATIVE INTERFACE MOCKUP

**SIT Policy Advisor** Ask

What are the exam regulations?

The answer is grounded in the uploaded policy documents.
[SIT_Student_Handbook_and_Exam_Regulations.pdf, p. 4]

▸ Transparency similarity 0.82

▸ Verbatim snippets

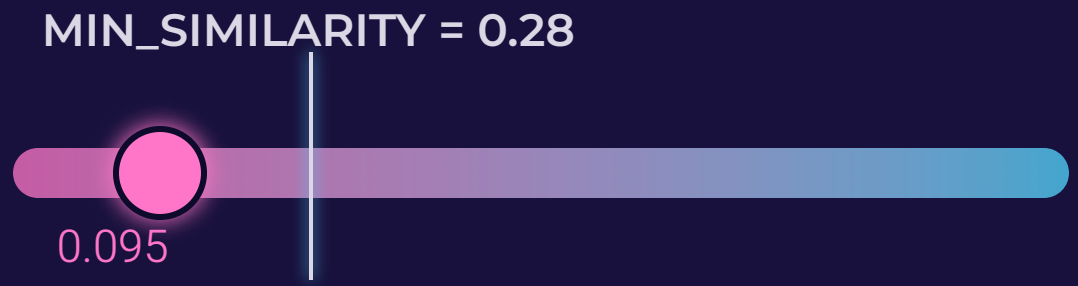
Illustrative interface mockup. The running Gradio UI is the intended live demonstration.

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HALLUCINATION CONTROL

# Low similarity triggers a safe refusal

The app bypasses Gemma when the evidence is too weak.




**Below the boundary**
The app bypasses Gemma.

**Refusal message**
“The answer could not be found in the documents.”

Example: “Who won the 2022 FIFA World Cup?” scores 0.095 and is refused.

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EVALUATION EVIDENCE

# 15 questions produced a clean evaluation result

## 100%

Overall accuracy across tested realistic similarity settings.

## 0

False positives: 0 hallucinated answers.

## 0

False negatives: 0 valid questions incorrectly refused.

Evaluation set: 14 valid policy questions plus 1 out-of-domain refusal question.

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THE PRODUCT PROMISE

# Reliability is the product feature

01

Dense local retrieval finds the right evidence.

02

Strict generation rules keep answers grounded.

03

Explicit citations and visible snippets make verification part of the experience.

04

The Gradio app is ready to index a new document set in seconds.

Q&A · Live demonstration

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